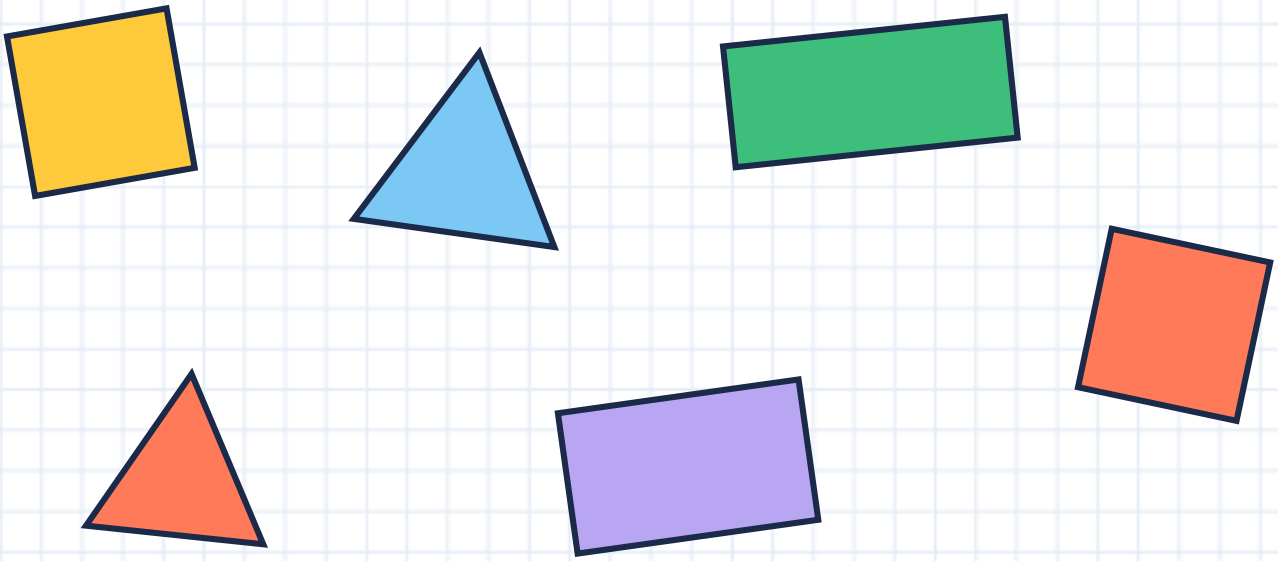


LEVEL 3 · HOLIDAY PROGRAM

Math Inquirers

Notice it. Test it. Explain why.

Name _____ **Group** _____

My Inquirer Rules

1. Notice the pattern.
2. Test it with objects first.
3. Check your answer.
4. Explain why.

My Inquiry Journal

Your tutor stamps each lesson.

Lesson 1
Number Sleuths

Lesson 2
Cryptarithm Crackers

Lesson 3
Times Table Treasure

Lesson 4
Model Makers

Lesson 5
Remainder Riddles

Lesson 6
Fraction Walls

Lesson 7
Area Architects

Lesson 8
Angle and Line Detectives

Lesson 9
Measure and Money Masters

Lesson 10
Challenge Day

Tutor: _____ Program dates: _____

NUMBER SLEUTHS

Name _____ Group _____ Date _____

Start here. Do page 1, then Q4 and Q5 on page 2. After that you may choose: Think harder, Go further or the Investigation.

Digit cards: use each card once in each number. A 4-digit number cannot start with 0.

No.1

Use these digit cards once each to make 4-digit numbers.

3	0	7	5

a The greatest number: _____

b The smallest number: _____

c The greatest ODD number: _____

d The number closest to 5000: _____

e Why is (d) the closest? It is only _____ away from 5000.

No.2

Make a list in order. Write ALL the 4-digit numbers you can make with the cards 2, 0, 5, 5. Start with the smallest.

Starting with 2	
Starting with 5	

a How many numbers did you find? _____

b How many of them are greater than 5200? _____

No.3

Look for a pattern. Write the next two numbers.

a 9850, 9750, 9650, _____, _____

b 1234, 2345, 3456, _____, _____

c 1000, 1001, 1003, 1006, 1010, _____, _____

d 3000, 3250, 3500, 3750, ... Keep going. Will 5000 be in the pattern? _____ Which number comes right after 5000? _____

No.4

Place value puzzles.

Hint: 10 hundreds make 1 thousand. You can draw a box for each thousand, hundred, ten and one.

a 4 thousands, 15 hundreds, 2 tens and 7 ones make _____

b 7305 = 6 thousands + _____ hundreds + 5 ones

c Mei says that 6 thousands, 0 hundreds and 23 tens is 6023. Is she right? Explain.

No.5

What number am I? Use one clue at a time.

a Read the clues.

Clue 1: I am between 6000 and 7000.
Clue 2: My ones digit is half of my thousands digit.
Clue 3: My hundreds digit is the sum of my other three digits.

I am: _____

b (if time) I am a 4-digit ODD number. All my digits are different. My digits add up to 6. What is the greatest number I can be? What is the smallest?

Greatest: _____
Smallest: _____

No.6

(if time) How many 4-digit numbers have digits that add up to 3? Make a list in order. Start with the biggest thousands digit.

Thousands digit 3	
Thousands digit 2	
Thousands digit 1	

a How many numbers altogether? _____

b How many start with 3? With 2? With 1? What do you notice?

Want more? You choose. Think harder is the next page. Or try the Investigation (What if one card changes?) on the last page of this lesson. Easy first step: with the cards 0, 1, 2, 3, write only the numbers that start with 1. Use the cards to help you.

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder is for anyone who wants it. Try a question, then try the Investigation on the last page of this lesson if you like.

Think harder

No.1

Use the cards 5, 0, 8, 2 once each. Make the greatest even number and the smallest odd number.

No.2

How many 4-digit numbers can you make with the cards 1, 1, 0, 0? List them.

No.3

A number has 7 thousands, 13 hundreds and 6 ones. What is the number?

No.4

I am between 5000 and 6000. My hundreds, tens and ones digits are all the same. My digits add up to 11. What number am I?

Go further

No.5

Make up your own "What number am I?" puzzle. Choose a secret 4-digit number and write 3 or 4 clues about its digits. Check that ONLY ONE number fits all your clues. Then swap with a partner.

No.6

A 4-digit number has all different digits. Its thousands digit is the sum of its other three digits. What is the greatest such number? What is the smallest?

No.7

Mei uses the cards 1, 2, 3, 4 once each to make 4-digit numbers. She lists them all from smallest to greatest. What is the 10th number in her list?

Story corner (optional): Rozsa Peter. Rózsa Péter (Hungary, 1905 to 1977) helped start the study of recursion: rules that build each number from the one before. In 1973 she became the first woman elected to the Hungarian Academy of Sciences. **Try it:** Start at 5. The rule is: double the number, then add 1. Write the next four numbers. Can you predict the one after that?

INVESTIGATION AND LOOK BACK

Name _____ Group _____ Date _____

Investigation: What if one card changes?

No.8

The cards 1, 2, 3, 4 make 24 four-digit numbers: 6 start with each card. Now change the 4 card to a 0, so the cards are 0, 1, 2, 3.

Easy first step: with the cards 0, 1, 2, 3, write only the numbers that start with 1. Use the cards to help you.

a Make a list in order of the numbers that start with 1. How many are there?

b How many start with 2? With 3? With 0?

c How many 4-digit numbers can you make altogether now? What stayed the same? What changed? Why?

Look back

Something I did well: ☐ I kept trying ☐ I asked a question ☐ I checked my answer ☐ I tried a new idea ☐ something else: _____

The hardest part was: ☐ making the list in order ☐ renaming (15 hundreds) ☐ working out the clues ☐ something else: _____

What helped me: ☐ drawing a picture ☐ making a list ☐ trying a number and checking ☐ I asked for help ☐ I am still stuck (I can ask for a check-in in the open class, with my parent nearby)

Two minutes. Tick or write. You may pass on any line. This page is yours: your tutor does not read it, read it out or collect it. Still stuck? Say "check-in please" in the open class, or ask your grown-up to tell your tutor.

CRYPTARITHM CRACKERS

Name _____ Group _____ Date _____

Start here. Do page 1, then Q4 and Q5 on page 2. After that you may choose: Think harder, Go further or the Investigation.

Work one column at a time, starting with the ones. Remember the carried 1!

No.1

Find the missing digits. Write them in the boxes. Example: 3 ___ 2 + 115 = 467. Ones: 2 + 5 = 7. Tens: ___ + 1 = 6, so the missing digit is 5.

(a)

$$\begin{array}{r} 5\square 2 \\ -\ 2\ 1\ 6 \\ \hline 3\ 1\ 6 \end{array}$$

(b)

$$\begin{array}{r} 3\square 4\ 7 \\ +\ 2\ 5\square 8 \\ \hline \square\ 0\ 2\ 5 \end{array}$$

c Hint for (a): turn it round and add. The answer plus the number taken away makes the first number. Check (a) by adding: 316 + 216 = _____

No.2

Use the digit cards 1 to 6, each once, to make two 3-digit numbers.

1	2	3	4	5	6

a Make the sum as GREAT as possible. _____ + _____ = _____

b Make the sum as SMALL as possible. _____ + _____ = _____

c Make the DIFFERENCE as small as possible. Guess, check, then improve. Write your tries.

.....

.....

Best: _____

No.3

Magic square: every row, column and diagonal adds up to the same total.

1200	1700	1600
	1500	

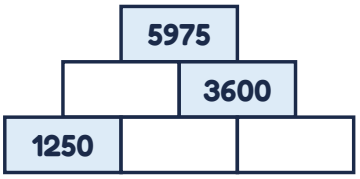
a What is the magic total? _____ Fill in the missing numbers.

b Look at two numbers opposite each other across the centre. What do you notice?

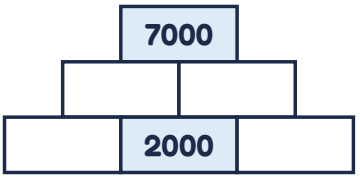
No.4

Number pyramids: each brick is the sum of the two bricks under it. Work backwards.

(a)



(b) corner bricks are equal



No.5

Look for a pattern.

a 4127 + _____ = 9999 and 2593 + _____ = 9999

b What do you notice about the digits? Why is there no carrying?

c Use your pattern: 10 000 – 4127 = _____

No.6

(if time) Letter puzzle: AB + AB + AB = CAB. Each letter stands for a digit. Different letters are different digits.

a A = _____ B = _____ C = _____

b The ones column: B + B + B ends in B. Which digits can B be? Try 0 and 5, then check the tens column. How do you know?

Want more? You choose. Think harder is the next page. Or try the Investigation (Is it always true? The reverse trick) on the last page of this lesson. Easy first step: try 842 – 248 = 594 for yourself. Then try 953 and 351 (reverse it, then subtract).

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder is for anyone who wants it. Try a question, then try the Investigation on the last page of this lesson if you like.

Think harder

No.1

Find the missing digits (turn it round and add): 7A03 – B46C = 2538

No.2

Find the missing digits: 2A7B + C6D9 = 6000

No.3

In a number pyramid, each brick is the sum of the two bricks under it. The bottom row is 1200, A, B. The middle row is 2650, C. The top brick is 6000. Find the missing bricks A, B and C.

No.4

Ali meant to add 1099 to a number, but he added 1999 by mistake and got 6543. What is the correct answer?

Go further

No.5

Make up your own missing-digits sum. Write a 4-digit + 4-digit addition that works, then rub out exactly 3 digits in 3 DIFFERENT columns and write the letters A, B and C instead. Swap with a partner. Does your partner find only one answer? Explain why your sum has only one answer.

No.6

Use the digit cards 1 to 8 once each to make two 4-digit numbers. Make their difference as small as possible.

No.7

ABC + ACB = CBA. Different letters are different digits. Find them.

Story corner (optional): Joan Clarke. Joan Clarke (England, 1917 to 1996) studied maths at Cambridge and worked as a codebreaker in Hut 8 at Bletchley Park. Codebreakers cut down how many possibilities they had to test, just as you do with letter puzzles. **Try it:** Each letter is a different digit. A + A + A = BA, and B is not 0. Which digits are A and B? Check by adding.

INVESTIGATION AND LOOK BACK

Name _____ Group _____ Date _____

Investigation: Is it always true? The reverse trick

No.8

Take a 3-digit number whose first digit is bigger than its last digit, like 842. Reverse it to make 248. Subtract the smaller from the bigger: 842 – 248 = 594.

Easy first step: try 842 – 248 = 594 for yourself. Then try 953 and 351 (reverse it, then subtract).

a Try 4 more numbers of your own. Write each subtraction and its answer.

b What do you notice about all your answers? Compare with a partner.

c Is it always true? Test it on a number that ends in 0, like 720, then explain why you think it works.

Look back

Something I did well: ☐ I kept trying ☐ I asked a question ☐ I checked my answer ☐ I tried a new idea ☐ something else: _____

The hardest part was: ☐ carrying ☐ finding missing digits ☐ magic squares and pyramids ☐ something else: _____

What helped me: ☐ drawing a picture ☐ making a list ☐ trying a number and checking ☐ I asked for help ☐ I am still stuck (I can ask for a check-in in the open class, with my parent nearby)

Two minutes. Tick or write. You may pass on any line. This page is yours: your tutor does not read it, read it out or collect it. Still stuck? Say "check-in please" in the open class, or ask your grown-up to tell your tutor.

TIMES TABLE TREASURE

Name _____ Group _____ Date _____

Start here. Do page 1, then Q4 and Q5 on page 2. After that you may choose: Think harder, Go further or the Investigation.

No.1

Patterns in the 9 times table. Fill in the answers.

9 ×	1	2	3	4	5	6	7	8	9	10
Answer	9	18								

a Look at the tens digits and the ones digits. Finish the sentences. The tens digits go _____. The ones digits go _____.

b Ali says 9 × 8 = 81, because 8 + 1 = 9. Use a pattern to explain why he is wrong.

No.2

Last digits in the 7 times table. Write only the LAST digit of each answer.

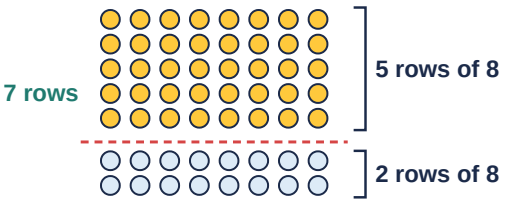
7 ×	1	2	3	4	5	6	7	8	9	10
Answer	7	14	21	28	35	42	49	56	63	70
Last digit	7	4								

a Test the 3, 6 and 9 times tables. Which of them also use all ten last digits? _____

b Without working out 7 × 46, what is its last digit? Explain.

No.3

Mei does not know 7 × 8. She splits 7 rows of 8 dots into two parts.



a 5 × 8 = _____ 2 × 8 = _____ so 7 × 8 = _____

b Split 9 × 6 into two facts you know. Show your working.

No.4

Grid puzzle. Each number in the grid is the row number × the column number. The row and column numbers are all from 1 to 10. Find the six missing row and column numbers, then fill in the grid.

×			
	56		
			81
	48	**36**	

Hint: 81 = 9 × 9 and 56 = 7 × 8.

No.5

Ben puts 36 chairs in equal rows (at least 2 rows, at least 2 chairs in each row). Find ALL the ways. Try 2 rows, then 3 rows, then 4 rows, and so on.

a Write only the ways that work. 3 rows of 12 is different from 12 rows of 3. Stop when you have all of them.

Number of rows	2							
Chairs in each row	18							

b How many ways are there? _____

c Most ways come in pairs, like 2 rows of 18 and 18 rows of 2. Which way has no partner? Why?

.....

.....

No.6

(if time) Number riddle.

a Find ALL the 2-digit numbers that can be divided exactly by 6 AND by 8. Hint: list the 2-digit numbers you can divide exactly by 8 (16, 24, 32 ...), then test each one.

.....

.....

b I am one of your numbers. My digits add up to 12. What number am I?

Answer: _____

Want more? You choose. Think harder is the next page. Or try the Investigation (Is it always true? Digit sums) on the last page of this lesson. Easy first step: write the 9 times table up to 9 × 6 and add the digits of each answer.

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder is for anyone who wants it. Try a question, then try the Investigation on the last page of this lesson if you like.

Think harder

No.1

The digits of every answer in the 9 times table (9 × 1 to 9 × 10) add up to 9. Which of these are in the 9 times table: 54, 65, 72, 83?

No.2

What is the last digit of 8 × 8 × 8? Use last digits only.

No.3

Ravi says: "Every answer in the 6 times table is even." Is he right? Explain why.

No.4

Find ALL the 2-digit numbers that can be divided exactly by 7 and whose digits add up to 10.

Go further

No.5

Make up your own riddle: "I am a 2-digit number in the ____ times table ..." Write 2 or 3 clues (about odd or even, the digits, being between two numbers). Check that only ONE number fits, by listing the table. Swap with a partner.

No.6

Ben keeps multiplying by 7: 7, 49, 343 ... He writes only the last digit each time: 7, 9, 3 ... What is the last digit when he has multiplied 8 sevens together? Explain the pattern.

No.7

I am a 3-digit number less than 200. I can be divided exactly by 7 and by 9. My digits add up to 9. What number am I?

Story corner (optional): John Napier. John Napier (Scotland, 1550 to 1617) looked after his family's farms, and maths was his hobby. He made rods with times tables written on them, called Napier's bones, so a big multiplication became reading rows and adding. **Try it:** Make a rod for 9. Write 9 x 1 up to 9 x 9. What pattern do you see in the tens digits and in the ones digits?

INVESTIGATION AND LOOK BACK

Name _____ Group _____ Date _____

Investigation: Is it always true? Digit sums

No.8

In the 9 times table the digits of each answer add up to 9: 9, 18 (1 + 8), 27 (2 + 7), 36 (3 + 6) ...

Easy first step: write the 9 times table up to 9 × 6 and add the digits of each answer.

a Carry on the 9 times table to 9 × 12. Add the digits of each answer. Is the total ALWAYS 9? Write what you find.

b Try the 3 times table and the 6 times table up to × 12. Add the digits of each answer. What do you notice?

c Try the 4 times table. Is it the same? What can you say about the 3, 6 and 9 times tables?

Look back

Something I did well: ☐ I kept trying ☐ I asked a question ☐ I checked my answer ☐ I tried a new idea ☐ something else: _____

The hardest part was: ☐ remembering the facts ☐ spotting the pattern ☐ explaining why ☐ something else: _____

What helped me: ☐ drawing a picture ☐ making a list ☐ trying a number and checking ☐ I asked for help

☐ I am still stuck (I can ask for a check-in in the open class, with my parent nearby)

Two minutes. Tick or write. You may pass on any line. This page is yours: your tutor does not read it, read it out or collect it. Still stuck? Say "check-in please" in the open class, or ask your grown-up to tell your tutor.

MODEL MAKERS

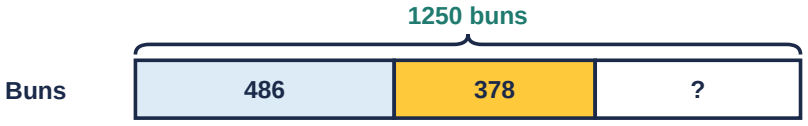
Name _____ Group _____ Date _____

Start here. Do page 1, then Q4 and Q5 on page 2. After that you may choose: Think harder, Go further or the Investigation.

Draw a model. Label what you know. Put a ? on what you want to find.

No.1

A bakery made 1250 buns. It sold 486 buns in the morning and 378 buns in the afternoon.



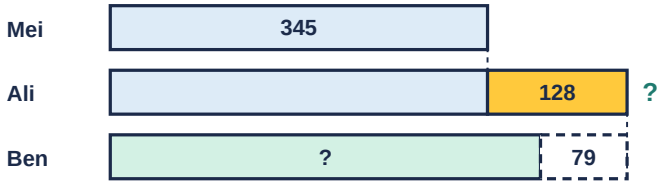
Hint: add the two amounts sold first (486 + 378). Then take that away from 1250.

a How many buns were left? _____

b How many more buns were sold in the morning than in the afternoon? _____

No.2

Mei has 345 stickers. Ali has 128 more stickers than Mei. Ben has 79 fewer stickers than Ali.



Hint: find Ali first (compare with Mei). Then find Ben (compare with Ali, not Mei).

a Ali has _____ stickers. Ben has _____ stickers.

b How many stickers do the three children have altogether? _____

No.3

Siti has 3 times as many beads as Ravi. They have 248 beads altogether. Each box in the picture is 1 unit (one equal part).



a How many units are there altogether? _____ 1 unit = _____ beads

b Siti has _____ beads. She has _____ more beads than Ravi.

No.4

Ali and Ben have 150 marbles altogether. Ali has 24 more marbles than Ben. How many marbles does each boy have?

Hint: you can guess and check. Try Ben = 50, so Ali = 74. Is the total 150? Too big or too small? Try again.

Draw your model here.

Ben: _____ Ali: _____

No.5

Mei has 85 stickers. Siti has 47 stickers. Mei gives some stickers to Siti so that they both have the same number.

Hint: try giving 10 stickers. Do they have the same number now? Keep trying.

Draw your model here.

a How many stickers does Mei give Siti? _____

b Ravi says, "Mei must give Siti 38 stickers, because 85 – 47 = 38." What mistake did Ravi make?

No.6

Ali has $45. Ben has $18. Grandma gives each boy $10.

Draw your model here.

a How much more money does Ali have than Ben now? _____

b How much more did Ali have than Ben before Grandma gave the money? _____

c What do you notice? Why?

Want more? You choose. Think harder is the next page. Or try the Investigation (What stays the same?) on the last page of this lesson. Easy first step: fill in the first row. In 1 year Mei is 14 and her brother is 6. What is the difference?

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder is for anyone who wants it. Try a question, then try the Investigation on the last page of this lesson if you like.

Think harder

No.1

Ali and Ben have 96 cards altogether. Ali has 3 times as many cards as Ben. How many more cards does Ali have than Ben?

No.2

Siti has 54 more beads than Mei. Siti gives Mei some beads, and now they have the same number. How many beads did Siti give Mei?

No.3

Three boxes hold 600 pens altogether. Box A has 50 more pens than Box B. Box C has 50 more pens than Box A. How many pens are in Box C?

No.4

A bag of rice and a bag of flour weigh 9 kg altogether. The rice is 3 kg heavier than the flour. How heavy is each bag?

Go further

No.5

Make up your own before-and-after story about stickers or money, where one child gives some to another child. Draw its model, solve it and check it. Then change ONE number in your story. What changes in the answer, and what stays the same? Swap with a partner.

No.6

Ravi is 7 years old and his sister is 3. In how many years will their ages add up to 30? Explain how you know.

No.7

A ribbon 4 m 75 cm long is cut into 3 pieces. The second piece is twice as long as the first. The third piece is 25 cm longer than the second. How long is the third piece?

Story corner (optional): Bhaskara II. Bhaskara II (India, 1114 to 1185) wrote the Lilavati, a maths book full of word problems about kings, elephants and pearl necklaces. A story says it was named after his daughter, but historians have no proof, so it is a legend. **Try it:** In the style of Bhaskara: a necklace has some pearls. Half of them fall on the floor, a quarter fall on the bed and 5 stay on the string. How many pearls were there? Draw a bar model.

INVESTIGATION AND LOOK BACK

Name _____ Group _____ Date _____

Investigation: What stays the same?

No.8

Mei is 13 years old and her brother is 5.

Easy first step: fill in the first row. In 1 year Mei is 14 and her brother is 6. What is the difference?

a Fill in the table for the next 5 years.

Years from now	Mei's age	Brother's age	Difference	Total of both ages
now				
1				
2				
3				
4				
5				

b What stays the same every year? What changes? How does it change?

c Will Mei ever be exactly DOUBLE her brother's age? When? What about TRIPLE? Explain.

Look back

Something I did well: ☐ I kept trying ☐ I asked a question ☐ I checked my answer ☐ I tried a new idea ☐ something else: _____

The hardest part was: ☐ drawing the bar model ☐ before and after stories ☐ finding what stays the same ☐ something else: _____

What helped me: ☐ drawing a picture ☐ making a list ☐ trying a number and checking ☐ I asked for help ☐ I am still stuck (I can ask for a check-in in the open class, with my parent nearby)

Two minutes. Tick or write. You may pass on any line. This page is yours: your tutor does not read it, read it out or collect it. Still stuck? Say "check-in please" in the open class, or ask your grown-up to tell your tutor.

REMAINDER RIDDLES

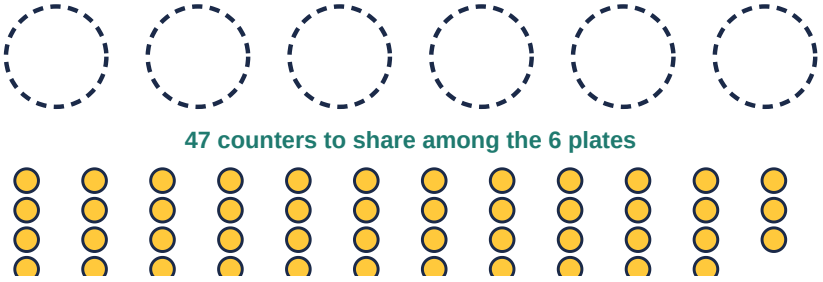
Name _____ Group _____ Date _____

Start here. Do page 1, then Q4 and Q5 on page 2. After that you may choose: Think harder, Go further or the Investigation.

When we share or make groups, some may be left over. The leftover is called the remainder. It is always smaller than the number we divide by.

No.1

Siti shares 47 stickers equally among 6 friends.



47 counters to share among the 6 plates

a How many stickers does each friend get? _____ How many are left over? _____

b What is the smallest number of extra stickers Siti needs so that none are left over? _____

c Ben works out 60 ÷ 7. He writes: 60 ÷ 7 = 7 remainder 11. What is wrong? Write the correct answer.

No.2

Divide. Then decide what to do with the leftover.

a 75 pupils. 1 minibus holds 9 pupils. How many minibuses? _____

b $60. 1 book costs $7. How many books can Mei buy? _____

c 87 eggs. 1 box holds 6 eggs. How many boxes are FULL? _____

d How many boxes hold ALL the eggs? _____

No.3

Ravi packs marbles into bags of 8. He fills 12 bags and has 5 marbles left over.

Hint: draw 12 bags of 8, then 5 more.

a How many marbles does Ravi have? _____

b (if time) He tips them all out and packs them again into bags of 9. How many full bags does he get? How many marbles are left over?

Full bags: _____ Left over: _____

No.4

What number am I? Make a list, then test each number.

a I am between 40 and 60. When I am divided by 5, the remainder is 3. When I am divided by 4, the remainder is 1.

Remainder 3 when ÷ 5				
Remainder when ÷ 4				

I am: _____

b (if time) I am a 2-digit number less than 50. When I am divided by 6, the remainder is 5. When I am divided by 4, the remainder is 3. Find ALL the numbers I could be.

.....

.....

No.5

Dividing by 8.

a Write ALL the remainders you could get when you divide a number by 8. Hint: share 9, 10, 11, ... counters into groups of 8. What is left over each time?

Answer: _____

b A number between 60 and 70 is divided by 8. The remainder is as big as it can be. What is the number? _____

No.6

(if time) At a class party, children sit at tables. If 6 children sit at each table, 5 children have no seat. If 7 children sit at each table, every seat is filled and nobody is left over.

Number of tables	1	2	3	4	5	6
Children: 6 at each table, 5 with no seat	11					
Children: 7 at each table	7					

a How many tables are there? _____ How many children? _____

b If 8 children sit at each table instead, how many seats are empty? _____

Want more? You choose. Think harder is the next page. Or try the Investigation (Is it always true? Three in a row) on the last page of this lesson. Easy first step: circle the number in the 3 times table in 14, 15, 16. Then try 20, 21, 22.

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder is for anyone who wants it. Try a question, then try the Investigation on the last page of this lesson if you like.

Think harder

No.1

Mei has 53 beads. She makes bracelets with 8 beads each. How many bracelets can she make? How many more beads does she need to make one more bracelet?

No.2

I am a number between 20 and 30. When I am divided by 4, the remainder is 3. When I am divided by 6, the remainder is 5. What number am I?

No.3

What is the biggest 2-digit number that leaves a remainder of 8 when it is divided by 9?

No.4

A boat can carry 9 people at a time. 100 people want to cross the river. How many boatloads are needed?

Go further

No.5

Make up your own remainder riddle: "I am a number between ____ and ____. When I am divided by ____ the remainder is ____. When I am divided by ____ the remainder is ____." Check that only ONE number fits, by listing the numbers for the first clue and testing them. Swap with a partner.

No.6

Today is Monday. What day of the week will it be 100 days from today? What day of the week was it 100 days ago? Explain how you know.

No.7

Mr Lim gives pens to his pupils. If each pupil gets 4 pens, 10 pens are left over. If each pupil gets 6 pens, he is 4 pens short. How many pupils are there? How many pens does he have?

Story corner (optional): Sun Zi. Almost nothing is known about Sun Zi (China, more than 1,500 years ago), but a Chinese maths book with his name, the Sunzi Suanjing, has a riddle about remainders. Its answer is 23: it leaves 2 in groups of 3, 3 in groups of 5 and 2 in groups of 7. **Try it:** Check the answer 23 for Sun Zi's riddle by dividing. Then find a number between 10 and 20 that leaves 1 when divided by 4 and leaves 2 when divided by 3. .....

INVESTIGATION AND LOOK BACK

Name _____ Group _____ Date _____

Investigation: Is it always true? Three in a row

No.8

Write any three whole numbers in a row, like 14, 15, 16. Exactly one of them is in the 3 times table (15).

Easy first step: circle the number in the 3 times table in 14, 15, 16. Then try 20, 21, 22.

a Try 5 different starting numbers (make one of them bigger than 100). Circle the number in the 3 times table each time. Is it always exactly one?

b Explain why. Think about the remainders when you divide by 3.

c What if we change it? Try 4 numbers in a row and the 4 times table. Then try only 2 numbers in a row and the 3 times table.

Look back

Something I did well: ☐ I kept trying ☐ I asked a question ☐ I checked my answer ☐ I tried a new idea ☐ something else: _____

The hardest part was: ☐ dividing ☐ reading the remainder ☐ riddles with two clues ☐ something else: _____

What helped me: ☐ drawing a picture ☐ making a list ☐ trying a number and checking ☐ I asked for help ☐ I am still stuck (I can ask for a check-in in the open class, with my parent nearby)

Two minutes. Tick or write. You may pass on any line. This page is yours: your tutor does not read it, read it out or collect it. Still stuck? Say "check-in please" in the open class, or ask your grown-up to tell your tutor.

FRACTION WALLS

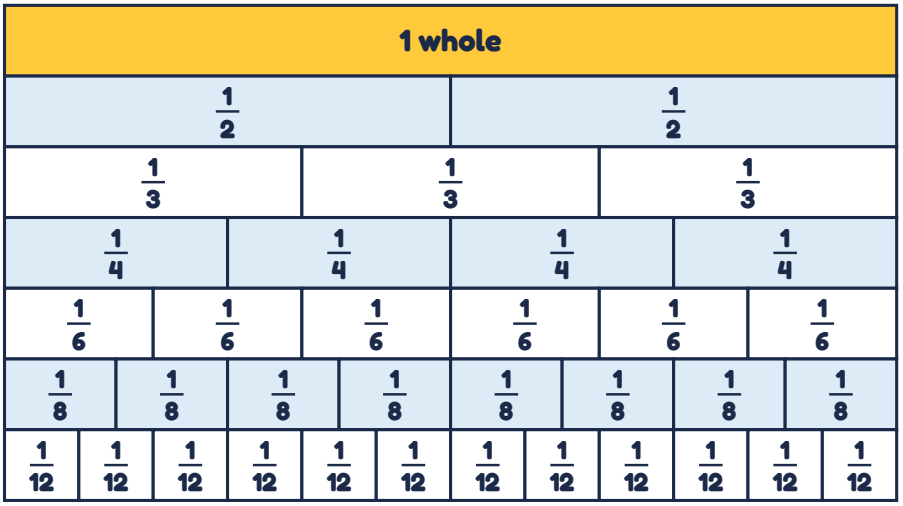
Name _____ Group _____ Date _____

Start here. Do page 1, then Q4 and Q5 on page 2. After that you may choose: Think harder, Go further or the Investigation.

Each row of a fraction wall is one whole, cut into EQUAL parts. To compare or add fractions, make the parts the same size.

No.1

Use the fraction wall.



a $\frac{1}{2} = \frac{\square}{4} = \frac{\square}{6} = \frac{\square}{8} = \frac{\square}{12}$

b $\frac{2}{3} = \frac{\square}{6} = \frac{\square}{12}$ (c) $\frac{3}{4} = \frac{\square}{8} = \frac{\square}{12}$

d Look at your answers to (a). What do you notice about the top and bottom numbers?

.....

No.2

Write >, < or = in each circle.

(a) $\frac{1}{5}$ ◯ $\frac{1}{7}$	(b) $\frac{3}{8}$ ◯ $\frac{5}{8}$	(c) $\frac{2}{3}$ ◯ $\frac{4}{6}$
(d) $\frac{3}{4}$ ◯ $\frac{5}{8}$	(e) $\frac{4}{9}$ ◯ $\frac{4}{7}$	(f) $\frac{2}{3}$ ◯ $\frac{3}{4}$

g Put in order, smallest first: $\frac{5}{12}$, $\frac{1}{2}$, $\frac{1}{3}$, $\frac{3}{4}$

Smallest first: _____

No.3

Each shape is cut into equal parts.



A B C D E

a Which shapes have exactly $\frac{1}{4}$ shaded? _____

b Write the fraction shaded in each of the other shapes. _____

No.4

Add or subtract. Make the parts the same size first.

(a) $\frac{2}{7} + \frac{3}{7} =$ _____	(b) $\frac{1}{4} + \frac{3}{8} =$ _____	(c) $\frac{1}{3} + \frac{5}{12} =$ _____
(d) $\frac{5}{6} - \frac{1}{3} =$ _____	(e) $1 - \frac{3}{10} =$ _____	(f) $\frac{1}{2} + \frac{3}{10} =$ _____

No.5

Ravi paints $\frac{1}{6}$ of a fence in the morning and $\frac{2}{3}$ of it in the afternoon. Draw a bar.

a What fraction has he painted? _____ (b) What fraction is not painted? _____

c How much more did he paint in the afternoon than in the morning? _____

No.6

Partners that make ONE WHOLE.

Pair each yellow card with the blue card that makes one whole with it. Write the 5 pairs.

$\frac{1}{4}$	$\frac{7}{12}$	$\frac{2}{3}$	$\frac{6}{8}$	$\frac{3}{8}$	$\frac{5}{6}$	$\frac{5}{12}$	$\frac{5}{8}$	$\frac{1}{6}$	$\frac{4}{12}$

Want more? You choose. Think harder is the next page. Or try the Investigation (Find all the ways) on the last page of this lesson. Easy first step: make 3/4 with one half. How much is left to fill?

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder is for anyone who wants it. Try a question, then try the Investigation on the last page of this lesson if you like.

Think harder

No.1

Find ALL the fractions with bottom number 12 that are bigger than 1/2 but smaller than 3/4. Hint: look at the twelfths row of your wall. Which one is equal to 2/3?

No.2

Which is bigger, 3/5 or 7/10? Show how you know.

No.3

Siti reads 1/4 of a book on Monday and 1/2 of the book on Tuesday. What fraction of the book has she read? What fraction is left?

No.4

Put these in order, smallest first: 2/3, 5/6, 7/12.

Go further

No.5

Make up your own "What fraction am I?" riddle, like: "I am equal to 3/4. My top and bottom numbers add up to ____." Choose the number so that exactly ONE fraction fits. Check it, then swap with a partner. Is your riddle always going to work for any number you choose? Explain.

No.6

Fill one whole bar using exactly 5 pieces. You may use 1/2, 1/4 and 1/8 pieces, as many of each as you like. Find all the ways.

No.7

Find a fraction that is bigger than 2/3 but smaller than 3/4. Hint: the fraction wall has no row that helps, so cut every part into smaller equal parts. Can you find another one?

Story corner (optional): Abu Bakr al-Hassar. Abu Bakr al-Hassar (Morocco and the Islamic West, 12th century; his exact dates are not known) wrote about fractions. He is described as the first known person to write the bottom number under a straight line and the top number above it, the fraction bar. **Try it:** Write two thirds and three quarters with a fraction bar. Use a fraction wall, or twelfths, to decide which one is bigger.

INVESTIGATION AND LOOK BACK

Name _____ Group _____ Date _____

Investigation: Find all the ways

No.8

Make exactly $\frac{3}{4}$ using pieces of $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{8}$. You may use as many of each as you like. Order does not matter ($\frac{1}{2} + \frac{1}{4}$ is the same way as $\frac{1}{4} + \frac{1}{2}$).

Easy first step: make 3/4 with one half. How much is left to fill?

a List every way. Start with the most halves.

b How do you know you have found them all?

c What if we want exactly 1 whole instead? Do you think there will be more ways or fewer? Test your idea.

Look back

Something I did well: ☐ I kept trying ☐ I asked a question ☐ I checked my answer ☐ I tried a new idea ☐ something else: _____

The hardest part was: ☐ equal fractions ☐ comparing fractions ☐ adding and taking away fractions ☐ something else: _____

What helped me: ☐ drawing a picture ☐ making a list ☐ trying a number and checking ☐ I asked for help ☐ I am still stuck (I can ask for a check-in in the open class, with my parent nearby)

Two minutes. Tick or write. You may pass on any line. This page is yours: your tutor does not read it, read it out or collect it. Still stuck? Say "check-in please" in the open class, or ask your grown-up to tell your tutor.

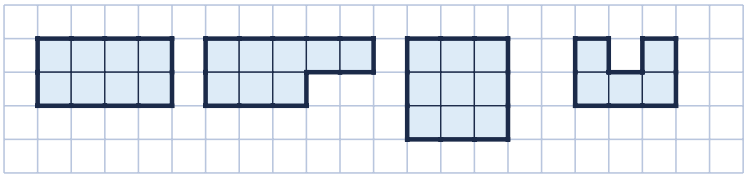
AREA ARCHITECTS

Name _____ Group _____ Date _____

Start here. Do page 1, then Q4 and Q5 on page 2. After that you may choose: Think harder, Go further or the Investigation.

No.1

Each small square is 1 cm by 1 cm. Find the area (in cm²) and the perimeter (in cm) of each figure.



A B C D

Figure	A	B	C	D
Area (cm²)				
Perimeter (cm)				

a Which two figures have the same area but different perimeters? _____

b Which figures have the same perimeter but different areas? _____

No.2

Siti has 18 m of fence. She makes a rectangular garden with the whole fence. Each side is a whole number of metres. List ALL the gardens she can make, in order.

Length (m)	Breadth (m)	Check: perimeter	Area (m²)

You may not need all the rows.

a Which garden has the biggest area? _____

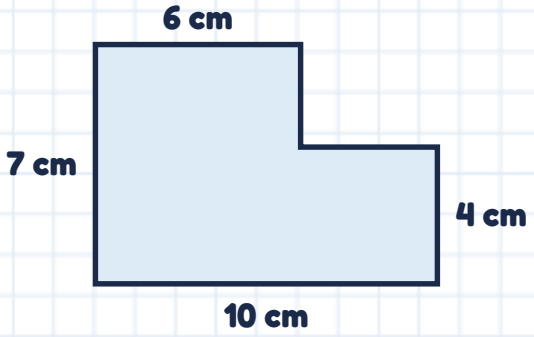
No.3

A square has the same perimeter as a rectangle 7 cm long and 3 cm wide. Hint: first walk round the rectangle. What is the area of the square? _____ cm²

No.4

Find the two missing sides first. Then find the area and the perimeter of this figure. Hint: you can cut the figure into two rectangles.

Missing sides: across = 10 – 6, down = 7 – 4.



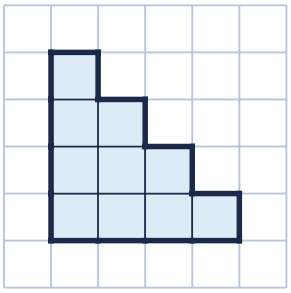
Missing sides: _____ and _____

a Area = _____ cm^2

b Perimeter = _____ cm

No.5

Look at this staircase.



a What is its area? _____ cm^2

b What is its perimeter? _____ cm

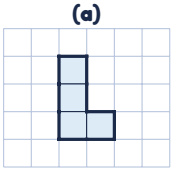
c Ben fills in the gaps with squares to make a 4 by 4 square. What happens to the perimeter? What happens to the area?

.....

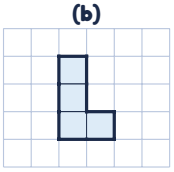
No.6

(if time) This L shape is made of 4 squares. Its perimeter is 10 cm. Add ONE square that shares a whole side with the L. Shade it.

(a)



(b)



a In grid (a), add a square so that the perimeter becomes 12 cm.

b In grid (b), add a square so that the perimeter stays 10 cm.

c Can you add one square so that the perimeter becomes 8 cm? Explain.

_____

Want more? You choose. Think harder is the next page. Or try the Investigation (What if we change it? What stays the same?) on the last page of this lesson. Easy first step: work out the area and perimeter of the 4 by 4 square and of the 5 by 3 rectangle.

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder is for anyone who wants it. Try a question, then try the Investigation on the last page of this lesson if you like.

Think harder

No.1

Find all the rectangles with whole-cm sides and an area of 24 cm². Which one has the shortest perimeter?

No.2

A rectangle is 12 cm long. Its perimeter is 30 cm. Find its area.

No.3

A rectangle 8 cm by 2 cm and a square have the same perimeter. Which has the bigger area, and by how much?

No.4

A square with sides of 6 cm is cut into two rectangles of the same size. What is the perimeter of each rectangle?

Go further

No.5

Make up your own rectangle puzzle. Choose a rectangle with whole-cm sides, then give a partner only TWO clues about it (for example its area and its perimeter). Check that only ONE rectangle fits both clues. Which two clues did you choose, and why do they work?

No.6

Ali joins 5 same-size squares with sides of 2 cm, whole sides touching. What is the biggest perimeter his shape can have? The smallest?

No.7

A rectangle 7 cm by 5 cm has a 1 cm square cut out of each of its 4 corners. Find the new area and the new perimeter.

Story corner (optional): Karen Uhlenbeck. Karen Uhlenbeck (USA, born 1942) studies minimal surfaces, the shapes of smallest area, like a soap film on a wire loop. In 2019 she became the first woman to win the Abel Prize, one of the top prizes in maths. **Try it:** Two wire frames have the same perimeter, 20 cm: a rectangle 6 cm by 4 cm, and a square 5 cm by 5 cm. Which soap film has the larger area? .....

INVESTIGATION AND LOOK BACK

Name _____ Group _____ Date _____

Investigation: What if we change it? What stays the same?

No.8

A square with sides 4 cm has area 16 cm² and perimeter 16 cm. Make it 1 cm longer one way and 1 cm shorter the other way. You get a 5 cm by 3 cm rectangle.

Easy first step: work out the area and perimeter of the 4 by 4 square and of the 5 by 3 rectangle.

a Work out the area and perimeter of the 5 by 3 rectangle. What changed? What stayed the same?

Square side	New rectangle	Perimeter of square	Perimeter of rectangle	Area of square	Area of rectangle
4 cm					
6 cm					
10 cm					
my own					

b Do the same for a 6 cm square (7 by 5) and a 10 cm square (11 by 9). Fill in the table. Then choose a square of your own.

c Is it always true? Draw the 4 by 4 square and explain what happens to the area.

Look back

Something I did well: ☐ I kept trying ☐ I asked a question ☐ I checked my answer ☐ I tried a new idea ☐ something else: _____

The hardest part was: ☐ finding missing sides ☐ listing all the rectangles ☐ telling area from perimeter ☐ something else: _____

What helped me: ☐ drawing a picture ☐ making a list ☐ trying a number and checking ☐ I asked for help ☐ I am still stuck (I can ask for a check-in in the open class, with my parent nearby)

Two minutes. Tick or write. You may pass on any line. This page is yours: your tutor does not read it, read it out or collect it. Still stuck? Say "check-in please" in the open class, or ask your grown-up to tell your tutor.

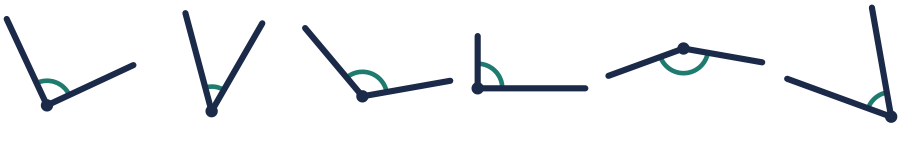
ANGLE AND LINE DETECTIVES

Name _____ Group _____ Date _____

Start here. Do page 1, then Q4 and Q5 on page 2. After that you may choose: Think harder, Go further or the Investigation.

No.1

Fold a right-angle checker. Use it to sort the angles a to f.

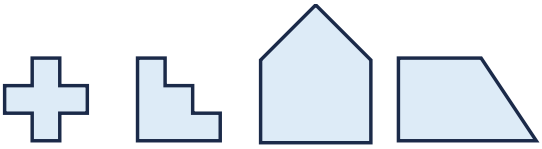


a b c d e f

Smaller: _____ Right angle: _____ Bigger: _____

No.2

How many right angles are INSIDE each shape? Check every corner with your checker. A corner that dents INTO the shape is bigger than a right angle: do not count it. A is done for you.

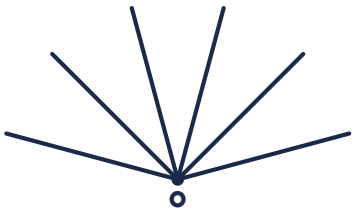


A B C D

Shape	A	B	C	D
Right angles inside	8			

No.3

These 6 lines all start at point O. (Example: 3 lines make 3 angles: 1 + 1 + 1.)



O

a How many angles? Count the small ones first, then those made of 2 small ones, and so on.

Made of 1: ____ 2: ____ 3: ____ 4: ____ 5: ____

Total: _____

b Look for a pattern. 8 lines start at one point, spread out like these. How many angles are there? _____

No.4

Look at the lines in these letters. Parallel lines never meet and stay the same distance apart, like train tracks. Perpendicular lines meet at a right angle (a square corner). Write Yes or No in the table.

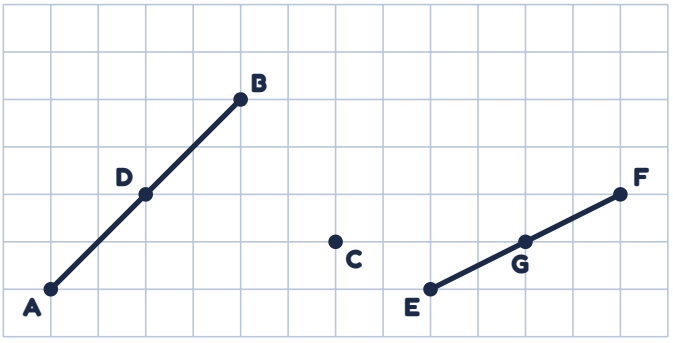
E H T L Z N

Letter	E	H	T	L	Z	N
Parallel lines?						
Perpendicular lines?						

b In the letter E: pairs of parallel lines _____ pairs of perpendicular lines _____

No.5

Use a ruler. Draw on the grid.

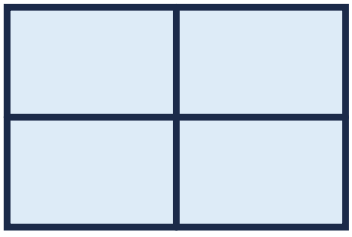


a Through C, draw a line parallel to AB. (b) Through D, draw a line perpendicular to AB.

c Think harder (if you have time): through G, draw a line perpendicular to EF. Check it with your checker. Hint: put your checker on EF first, then slide it to G.

No.6

How many right angles can you find in this window?



Visit every point where lines meet. Write how many right angles are at each point, then add.

Right angles altogether: _____

Want more? You choose. Think harder is the next page. Or try the Investigation (Staircase corners: find the pattern) on the last page of this lesson. Easy first step: draw the 1-step and 2-step staircases. Count the corners of the 2-step staircase.

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder is for anyone who wants it. Try a question, then try the Investigation on the last page of this lesson if you like.

Think harder

No.1

I have 4 sides and 4 right angles. My sides are not all the same length. What shape am I? How many pairs of parallel lines do I have?

No.2

A U shape is made of 5 squares: 3 squares in a row with 1 more square on top of each end square. How many right angles are inside it?

No.3

Look at a clock face. For each o'clock time from 1 o'clock to 11 o'clock, is the angle between the hands (the short way round) smaller than, the same as, or bigger than a right angle? Put each time in the right group.

No.4

The # sign is made of 4 lines: 2 upright lines and 2 lines across. How many pairs of parallel lines does it have? How many pairs of perpendicular lines?

Go further

No.5

Make up your own "Which shape am I?" riddle about a shape on square grid paper. Use words like sides, right angles, parallel and perpendicular, and give 3 clues. Draw the answer. Then check: can a partner draw a DIFFERENT shape that fits your clues? If yes, add a clue.

No.6

A long rectangle is cut into 6 squares in a row by 5 lines. How many right angles can you find?

No.7

Can a 6-sided shape have 6 right angles inside it? What is the most right angles a 6-sided shape can have inside?

Story corner (optional): Ibn al-Haytham. Ibn al-Haytham (about 965 to about 1040, born probably in Basra, worked in Cairo) wrote the Book of Optics about light and sight, and tested his ideas with experiments. He used geometry to work out where light goes when it bounces off a mirror. **Try it:** A ray of light hits a flat mirror and makes an angle of 30 degrees with the mirror. The perpendicular to the mirror makes 90 degrees with it. What angle does the ray make with the perpendicular? The reflected ray leaves at the same angle on the other side of the perpendicular. How big is the angle between the two rays?

INVESTIGATION AND LOOK BACK

Name _____ Group _____ Date _____

Investigation: Staircase corners: find the pattern

No.8

A 1-step staircase is 1 square. A 2-step staircase has 3 squares (a column of 2 beside a column of 1). A 3-step staircase has 6 squares.

Easy first step: draw the 1-step and 2-step staircases. Count the corners of the 2-step staircase.



a Count the corners of each staircase where the inside angle is a right angle. Then count the corners where the inside angle is BIGGER than a right angle. Draw the 4-step staircase and fill in the table.

Steps	Right-angle corners	Corners bigger than a right angle	All corners
1			
2			
3			
4			
10 (predict)			

b What pattern do you see? Use it to fill in the 10-step row without drawing.

c Is the number of corners always an even number? Why do you think so?

.....

.....

.....

Look back

Something I did well: ☐ I kept trying ☐ I asked a question ☐ I checked my answer ☐ I tried a new idea ☐ something else: _____

The hardest part was: ☐ counting angles ☐ parallel or perpendicular ☐ using the right-angle checker ☐ something else: _____

What helped me: ☐ drawing a picture ☐ making a list ☐ trying a number and checking ☐ I asked for help

☐ I am still stuck (I can ask for a check-in in the open class, with my parent nearby)

Two minutes. Tick or write. You may pass on any line. This page is yours: your tutor does not read it, read it out or collect it. Still stuck? Say "check-in please" in the open class, or ask your grown-up to tell your tutor.

MEASURE AND MONEY MASTERS

Name _____ Group _____ Date _____

Start here. Do page 1, then Q4 and Q5 on page 2. After that you may choose: Think harder, Go further or the Investigation.

No.1

Trick 1: change to ONE unit. 1 m = 100 cm 1 kg = 1000 g 1 ℓ = 1000 ml $1 = 100¢

a 2 m = _____ cm and 3 kg = _____ g

b 3 m 6 cm = _____ cm and 2 kg 450 g = _____ g

Hint for (b): write 3 m as 300 cm first.

c 4080 ml = _____ ℓ _____ ml

d Mei cuts 4 pieces, each 65 cm long, from a ribbon 3 m long. How much ribbon is left?

_____ cm

No.2

Four bags of rice are weighed: 1 kg 300 g 1030 g 1 kg 1 kg 3 g

a Order them, lightest first: _____, _____, _____, _____

b Ben says, "1 kg 30 g is the same as 130 g." Explain his mistake.

.....

No.3

Siti has $20. She wants to buy 3 different things from this list.

Book	Pen set	Puzzle	Stickers	Water bottle
$7.85	$4.60	$6.90	$2.35	$5.50

a Can she buy the book, the puzzle and the water bottle? Show how you know.

_____

.....

No.4

The 24-hour clock. After 12 noon, add 12 to the hour.

7:15 a.m.	3:40 p.m.	_____	12:30 p.m.
_____	_____	**20:05**	_____

a Fill in the table. The top row is 12-hour time; the bottom row is 24-hour time.

b Siti says, "17:00 is 7 p.m." Is she right? Explain.

No.5

Trick 2: draw a time line and jump to the next o'clock.

a Ali's swimming lesson starts at 3:45 p.m. and ends at 5:10 p.m. How long is it? _____ h _____ min

b Mei's piano lesson starts at 4:35 p.m. and lasts 50 minutes. When does it end? _____

c A film lasts 1 h 50 min. It ends at 9:15 p.m. When did it start? _____

Want more? You choose. Think harder is the next page. Or try the Investigation (Find all the ways to pay) on the last page of this lesson. Easy first step: pay 50 cents with ten 5-cent coins. Then swap some of them for 10-cent coins.

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder is for anyone who wants it. Try a question, then try the Investigation on the last page of this lesson if you like.

Think harder

No.1

A 2 ℓ bottle of juice is poured into cups. Each cup holds 300 ml. How many cups can be filled? How much juice is left over?

No.2

A shop opens at 09:30 and closes at 21:00. How long is it open?

No.3

A box with 4 books weighs 1 kg 50 g. The empty box weighs 250 g. All the books weigh the same. How heavy is one book?

No.4

Buses leave the interchange every 25 minutes. The first bus leaves at 06:40. The ride to school takes 35 minutes. Ravi must reach school by 08:30. Which is the latest bus he can take?

Go further

No.5

A bottle full of water weighs 1 kg 200 g. After half of the water is poured out, it weighs 750 g. How heavy is the empty bottle?

No.6

Mei buys 3 pens and 2 files for $9.20. Each file costs 60¢ more than each pen. How much does one pen cost?

No.7

Make up your own problem about money or time that needs at least 2 steps. Write it, solve it and check it. Then change ONE number and solve it again. What changes in the answer, and why? Swap with a partner.

Story corner (optional): Zu Chongzhi. Zu Chongzhi (China, about 1,500 years ago; sources differ on his exact dates) made a new calendar and measured the year as 365.24281 days, within about a minute of what astronomers find today. He also found that pi lies between 3.1415926 and 3.1415927. **Try it:** Zu's year was about 365 and a quarter days. How many hours is a quarter of a day? How many days are there in 4 years of that length? .....

INVESTIGATION AND LOOK BACK

Name _____ Group _____ Date _____

Investigation: Find all the ways to pay

No.8

Ali pays exactly 50 cents using only 5-cent, 10-cent and 20-cent coins. He can use as many of each coin as he likes.

Easy first step: pay 50 cents with ten 5-cent coins. Then swap some of them for 10-cent coins.

a Find all the ways. Write each way in the table (how many of each coin).

20-cent coins	10-cent coins	5-cent coins

b Which way uses the FEWEST coins? Which uses the MOST?

c How do you know you have found them all?

.....

.....

Look back

Something I did well: ☐ I kept trying ☐ I asked a question ☐ I checked my answer ☐ I tried a new idea ☐ something else: _____

The hardest part was: ☐ changing units ☐ working with money ☐ the 24-hour clock ☐ something else: _____

What helped me: ☐ drawing a picture ☐ making a list ☐ trying a number and checking ☐ I asked for help

☐ I am still stuck (I can ask for a check-in in the open class, with my parent nearby)

Two minutes. Tick or write. You may pass on any line. This page is yours: your tutor does not read it, read it out or collect it. Still stuck? Say "check-in please" in the open class, or ask your grown-up to tell your tutor.

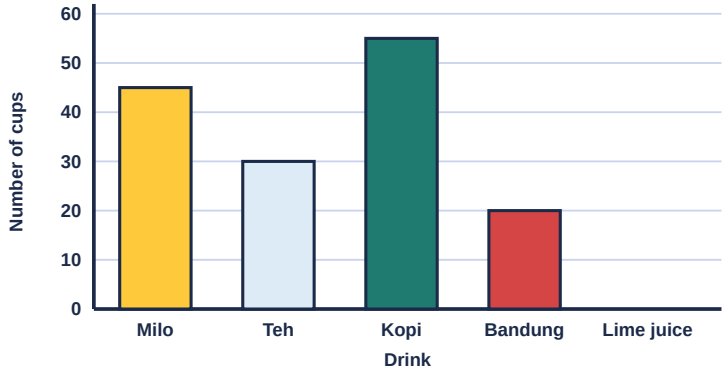
CHALLENGE DAY

Name _____ Group _____ Date _____

Start here. Do page 1, then Q4 and Q5 on page 2. After that you may choose: Think harder, Go further or the Investigation.

No.1

A drinks stall sold 200 cups of drinks on Monday. The bar graph shows the cups sold, but the bar for Lime juice is missing.



a Which drink shown sold the most? _____ (b) How many more cups of Kopi than Teh? _____

c How many cups of Lime juice were sold? _____ Now draw its bar on the graph.

d Which two drinks together sold exactly 100 cups? _____ and _____

No.2

Ali, Mei, Ravi and Siti are each in a different CCA. Use the clues: put a ✓ or a cross in each box, then write each child's CCA in the last column.

Clue 1 Ali is not in the choir and does not play chess.

Clue 2 Mei and Ravi are not in the choir.

Clue 3 The swimmer's name has 4 letters.

	Chess	Choir	Football	Swimming	CCA
Ali					
Mei					
Ravi					
Siti					

No.3

Draw it! Queues.

a Ben is 3rd from the front and 5th from the back. How many in the queue? _____

b Ali is 2nd from the front, Siti is last, 3 people are between them. How many in all? _____

No.4

List it! Make 3-digit numbers with these cards. Use each card once in every number.

2	5	8

a List ALL the numbers, in order from smallest to largest.

.....

b How many of them are odd numbers? _____

c Which number is closest to 600? _____

No.5

Make a table! Squares are made in a row with sticks. Fill in the table.



square: 4 sticks 2 squares: 7 sticks 3 squares: 10 sticks

Squares	1	2	3	4	5	6
Sticks	4	7	10			

a How many sticks are needed for 8 squares? _____

b Ben uses 31 sticks. How many squares does he make? _____

No.6

(if time) I am a 3-digit odd number. My digits add up to 6. My hundreds digit is bigger than my tens digit. None of my digits is 0. What could I be? Find ALL the answers.

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Want more? You choose. Think harder is the next page. Or try the Investigation (Handshakes: draw it, list it, make a table) on the last page of this lesson. Easy first step: draw 3 dots and join every pair. Count the lines. Then try 4 dots.

THINK HARDER

Name _____ Group _____ Date _____

Draw a model, make a list or make a table. Then write how you know.

Think harder is for anyone who wants it. Try a question, then try the Investigation on the last page of this lesson if you like.

Think harder

No.1

Ali has three coins: 10¢, 20¢ and 50¢. He picks one coin, or two coins, or all three. List every different amount of money he can make, in order. How many amounts are there?

No.2

Ali, Ben and Siti each have a different pet: a cat, a dog and a fish. Ali's pet has no fur. Ben does not have the dog. Who has which pet?

No.3

Triangles are made in a row with sticks: 1 triangle uses 3 sticks, 2 triangles use 5 sticks, 3 triangles use 7 sticks. How many sticks are needed for 12 triangles?

No.4

Use the digits 3, 0, 6 and 9 once each. Make the largest even 4-digit number and the smallest odd 4-digit number.

Go further

No.5

Make up your own logic puzzle with 3 people and 3 things (colours, pets or sports). Write 3 or 4 clues. Test it on a partner: does it have exactly ONE answer? Is there a clue you could take away and still have only one answer?

No.6

In a queue, Siti is 6th from the front and Ravi is 4th from the back. There are 3 children between them, and Siti is in front of Ravi. How many children are in the queue?

No.7

A bar graph shows the paper collected by 4 classes. 3P1 collected 10 kg more than 3P2. 3P3 collected the same as 3P1. 3P4 collected 5 kg less than 3P2. Altogether they collected 135 kg. How much did each class collect?

Story corner (optional): Raymond Smullyan. Raymond Smullyan (USA, 1919 to 2017) was a mathematician, logician, magician and pianist who loved puzzles. In his famous puzzles a knight always tells the truth and a knave always lies, and you use the clues to work out who is who. **Try it:** Ana says: 'Ben is a knave.' Ben says: 'We are both knights.' A knight always tells the truth and a knave always lies. Who is the knight and who is the knave? .....

INVESTIGATION AND LOOK BACK

Name _____ Group _____ Date _____

Investigation: Handshakes: draw it, list it, make a table

No.8

Some friends meet. Everyone shakes hands with every other friend once. Two friends make 1 handshake.

Easy first step: draw 3 dots and join every pair. Count the lines. Then try 4 dots.

a Find the number of handshakes for 3, 4, 5 and 6 friends. Draw dots and lines, or make a list, to help. Fill in the table.

Friends	Handshakes	How many more than the row above?
2		
3		
4		
5		
6		
10 (predict)		

b What pattern do you see? Use it to find the handshakes for 10 friends.

c Could there ever be exactly 20 handshakes? Explain.

Look back

Something I did well: ☐ I kept trying ☐ I asked a question ☐ I checked my answer ☐ I tried a new idea ☐ something else: _____

The hardest part was: ☐ choosing how to start ☐ the logic grid ☐ reading the bar graph ☐ something else: _____

What helped me: ☐ drawing a picture ☐ making a list ☐ trying a number and checking ☐ I asked for help

☐ I am still stuck (I can ask for a check-in in the open class, with my parent nearby)

Two minutes. Tick or write. You may pass on any line. This page is yours: your tutor does not read it, read it out or collect it. Still stuck? Say "check-in please" in the open class, or ask your grown-up to tell your tutor.

Cut-outs

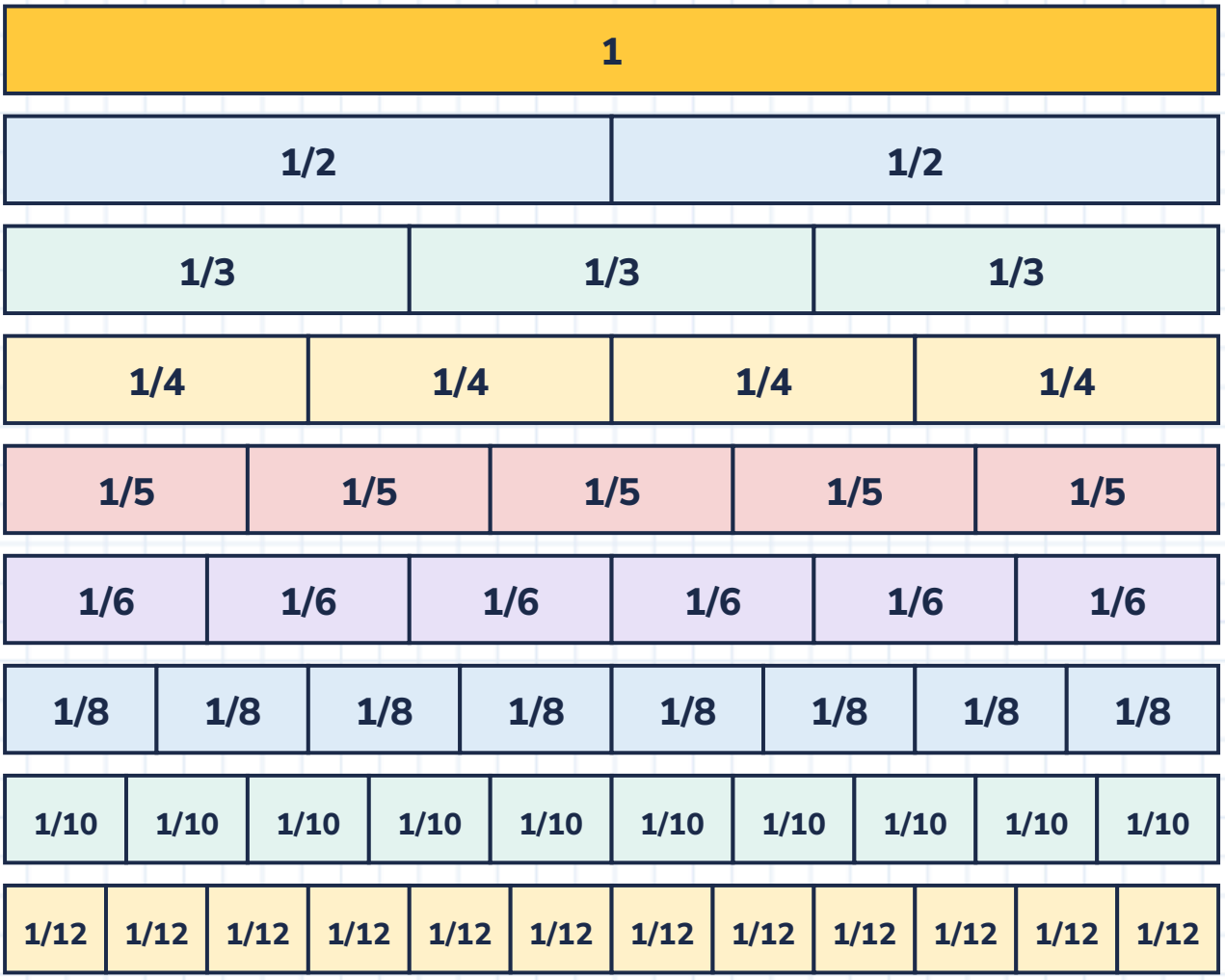
For the Level 3 program. Cut along the dashed lines and keep the pieces in one envelope.

Digit cards (Lessons 1 and 2)

0	1	2	3	4
5	6	7	8	9
0	1	2	3	4
5	6	7	8	9

Fraction wall

For Lesson 6. Cut the strips apart.

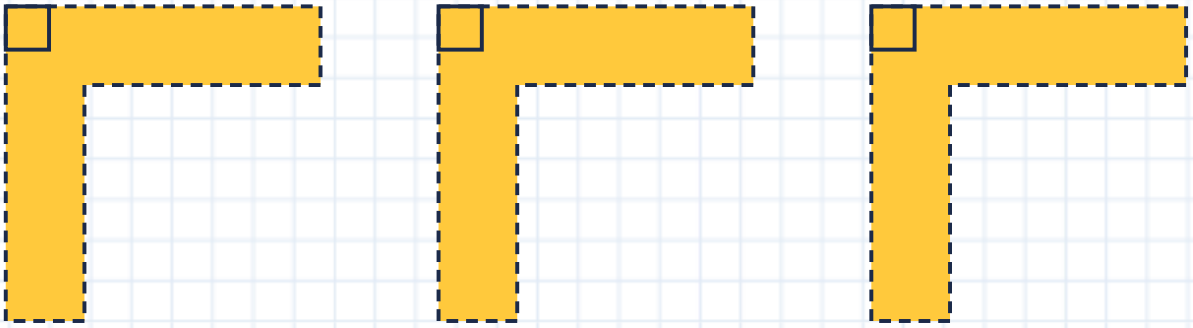


Line the strips up under the whole. Which fractions are the same size?

Square grid

For Lesson 7. Each square is one unit.

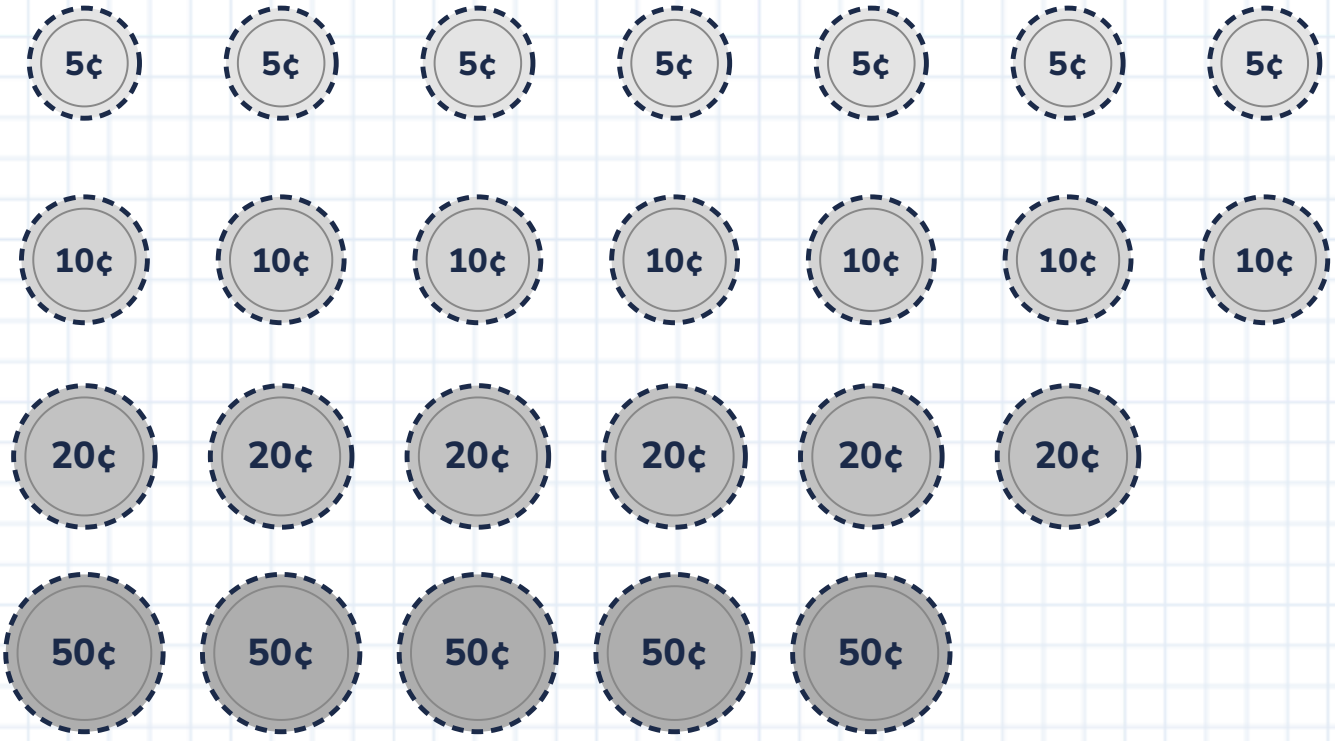
Right-angle testers (Lesson 8)



Cut out a right-angle tester. Put its corner on an angle: bigger, smaller or exactly a right angle?

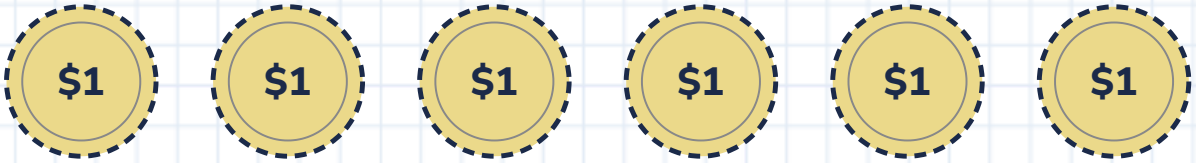
Play coins

For Lesson 9.



Play notes and $1 coins

For Lesson 9.





CERTIFICATE OF COMPLETION

Math Inquirers Holiday Program

Level 3 Holiday Program

This certifies that

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completed all 10 lessons, solved puzzles, spotted patterns and explained their thinking.

Notice it. Test it. Explain why.

Tutor: _____ Date: _____